Task 04: Evaluate the Services an Operating System Must Provide

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 4 Marks (Bloom's: K5 - Evaluate)
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> ***Task 4 Description:*** *Evaluate OS services across UI, program execution, I/O, file systems, communication, error detection, resource allocation, accounting, and protection. Determine services to remove from bedside monitors. Identify essential database services and patient record protection mechanisms. Evaluate desktop OS on tablets.*

1. SERVICE INCLUSION & EXCLUSION MATRIX

OS Service	Bedside Monitor (Striped RTOS)	Central Database Server	Engineering Justification
User Interface	**Minimal/Omitted:** Direct static frame-buffer writing; no window server or shell.	**Omitted:** Headless execution; remote management via SSH and API calls.	Eliminating GUI daemons saves RAM and prevents graphics crashes from affecting core services.
Program Execution	**Static:** Fixed set of pre-compiled firmware tasks. Dynamic loading is disabled.	**Dynamic:** Concurrent multi-process and multi-threaded query pools.	Preventing runtime binary loading on monitors eliminates memory fragmentation and injection vulnerabilities.
File System	**Omitted:** In-memory circular buffers; no disk journaling or block storage drivers.	**Critical:** ACID-compliant journaling file systems (XFS/ext4) and raw NVMe block devices.	Disabling persistent file writes on monitors avoids flash wear and write latency spikes.
I/O Operations	**Direct / Dedicated:** Direct memory-mapped I/O to sensor ADCs and GPIO alarms.	**High-Throughput:** Asynchronous I/O via `io_uring` and `epoll`.	Deterministic sensor readings require immediate I/O response without kernel buffering delays.
Accounting	**Omitted:** No user session tracking or CPU usage metrics.	**Strict:** Audit logs of queries, database access, and user identity tracking.	Eliminates unnecessary logging overhead on real-time microcontrollers.
Protection / Security	**Hardware MPU:** Read-only firmware regions and task memory segment isolation.	**Mandatory Access Control:** SELinux policies, TLS 1.3, disk encryption at rest.	Protects patient records from unauthorized access and prevents cross-tenant data leaks.

2. PATIENT RECORD MEMORY PROTECTION MECHANISMS

The database OS enforces record isolation across multiple levels:

- **Hardware Memory Translation (Paging):** The MMU maps virtual memory addresses to physical frames. Each database query process runs with a distinct base page table pointer (`CR3` register on x86_64). Process memory spaces are isolated at the hardware level; Doctor A's process cannot address Doctor B's memory frames.
- **Kernel Mandatory Access Control (MAC):** SELinux enforces strict least-privilege labels on file descriptors, sockets, and memory segments, preventing unauthorized process access even if root privileges are compromised.

3. ALTERNATIVE CONCEPT: FULL DESKTOP OS ON HANDHELD TABLETS

- **Space Overhead:** A desktop OS installs background printing spoolers, accessibility daemons, and desktop shells, requiring > 20 GB disk storage and 2–3 GB idle RAM, compared to ~500 MB for a tailored mobile OS.
- **Time & Battery Impact:** Desktop kernels lack aggressive wake-lock duty-cycling. Active background polling reduces battery runtime from 14 hours down to ~3.5 hours, failing to cover a 12-hour nursing shift. Boot times increase from 12 seconds to over 60 seconds.